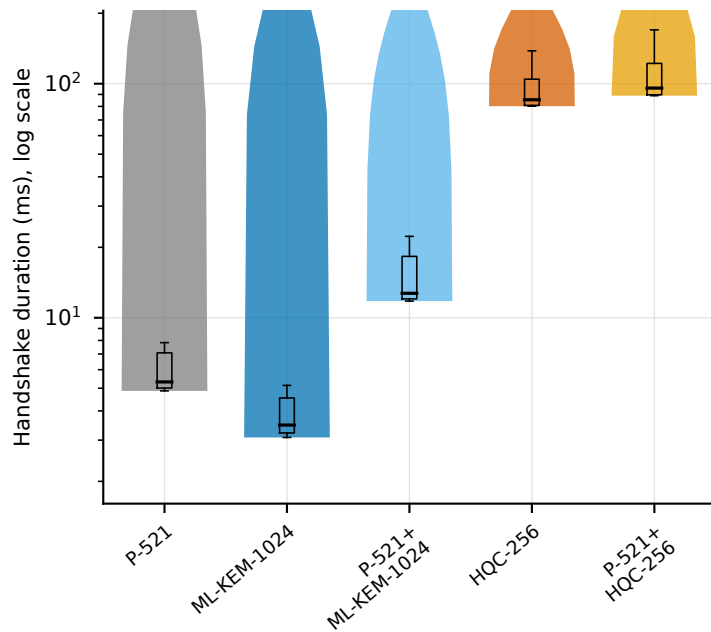
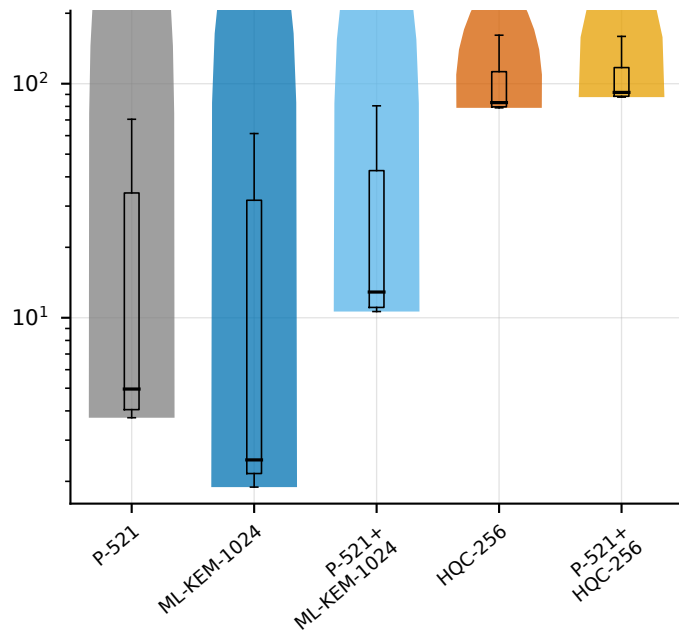


(a)

secp521r1 (classical)

**(b)**

mlds87 (post-quantum)

**(c)**
 Classical KEX
 ML-KEM (pure)
 ML-KEM (hybrid)
 HQC (pure)
 HQC (hybrid)

KEM	secp521r1 Mean (ms)	secp521r1 CV (%)	secp521r1 Max (ms)	secp521r1 Fail	mlds87 Mean (ms)	mlds87 CV (%)	mlds87 Max (ms)	mlds87 Fail
P-521	326.7	275.3	7.00k	2/500	719.4	225.3	7.00k	2/500
ML-KEM-1024	269.0	246.7	7.00k	0	540.1	239.3	8.03k	2/500
P-521+ML-KEM-1024	232.7	218.8	3.03k	0	477.8	243.8	7.14k	1/500
HQC-256	163.3	184.1	3.08k	0	204.8	187.1	3.10k	0
P-521+HQC-256	221.8	214.7	7.05k	0	260.6	215.2	7.06k	0